

AI Evals

Everything Module 7 taught, on one page. The research numbers will move; the shapes will not: coverage beats volume, binary beats a scale, read to the first failure, and an unchecked judge is an automated opinion.

1 · WHAT AN EVAL IS — LESSON 7.1 · YOUR BAR, WRITTEN DOWN BEFORE THE OUTPUT, IN A FORM THAT RE-RUNS

PIECE 1 · 7.2
The examples
Chosen, not a log dump

PIECE 2 · 7.3
The bar
Good, written down

PIECE 3 · 7.6
The score
Same way, every time

THE SPOT CHECK IS NOT EVIDENCE
1 in 10 bad, you read 10: **0.9¹⁰ ≈ 35%** chance you see nothing.
Meanwhile **40,000** requests shipped.

Without it, **“is it good”** is settled by the most senior opinion in the room, which does not survive a model swap. **WHY YOU CARE:** an engineer can build any bar you name. Nobody else is paid to name it.

157 ORGANISATIONS · JUNE 2026
50% of teams running evals shipped an agent that passed them, then failed a customer. Only **5%** fully trust automated evaluation.

2 · THE GOLDEN SET — LESSON 7.2

Coverage beats volume. Ten thousand logged conversations prove the easy case is easy; twelve chosen hard ones teach you something. Take the log file. Do not make it the set.

THE INPUT	WHAT GOOD LOOKS LIKE	WHY IT IS HERE
Verbatim, plus state	A condition, not a string	The failure it guards

Sources: **real traffic**, read for the odd ones · **what you fear** (needs no users) · **synthetic**, fills gaps, never the set.

Size: read **30 yourself** first, work toward **~100**, stop at **theoretical saturation** (no new failure types), then **10–20 a week**. A judge you trust needs **100–200 labelled per failure mode**. Every production failure becomes a row, permanently. **WHY YOU CARE:** which situations belong in the set is a product call.

4 · READING TRACES — LESSON 7.4

Trace: the full record of one request, one shared ID. **Span:** one step in it (model asked and answered, tokens, duration, why it stopped).

Go down until the first thing that went wrong. Then stop. Downstream, the model is working perfectly on rubbish. **“The model hallucinated”** was a calendar fetch back in **9 ms, empty**. An afternoon, not a new model. Read the same waterfall again for **slow**: four queued fetches in parallel, 800 → 200 ms.

Fifty traces → a work list: **notes**, then **names**, then **counts**. Categories first only finds what you suspected. Ask any tool: **does it emit OpenTelemetry?** **WHY YOU CARE:** the summary is where the next failure hides.

6 · MAKING IT RUN WITHOUT YOU — LESSON 7.6 · THREE EVALUATORS, CHEAPEST FIRST

1 · START HERE · FREE, NEVER WRONG
A plain check in code. Lookups: rows 1, 2, 5. Most rubrics are over half lookups.

2 · WHAT IS LEFT · CHEAP, APPROXIMATE
A model as judge. The row, its two examples, the output. Verdict + reason.

3 · THE THIN SLICE · THE REFERENCE
A person. What the other two cannot settle, plus a standing sample.

Good half: a strong judge agrees with humans **~80%** on well-structured tasks, about as often as two humans do. **Bad half:** 2026 bias tests trip frontier judges **over half the time**; verdicts move on rewording and **answer order**. **Check it:** grade 50 yourself, compare **passes and fails separately**, keep 20–30 a month hand-graded, recalibrate past **~20–25%** drift.

ANY AI QUALITY REVIEW — FIVE QUESTIONS TO ASK

- Who wrote down what good means, and when?** No date: seniority is the rubric.
- Where did the examples come from?** A log dump measures traffic.
- Which rows are code, which a judge, and when did a person last check it?**
- What used to pass and now fails?** The headline hides it.
- Offline number, online number, and the gap?** Big gap: feed failures in.

3 · THE RUBRIC — LESSON 7.3

Finished when two people using it reach the same verdict, alone. Six in ten on a first try is normal; a coin flip gets five. Stop describing a quality, **name what must be present**.

- One question per row.** Contains “and”? Two rows.
- Binary, not 1 to 5.** A real middle: fail / partial / pass, with a reason.
- A pass and a fail example under every row.** Later, the judge’s input.
- Your product’s words**, not “faithfulness” or “groundedness”.

One owner decides (the benevolent dictator) and writes each ruling in. Five experts to judge one interaction is a **scope** problem. **WHY YOU CARE:** that owner is you.

1	Every proposed slot falls inside all four attendees’ working hours.	CODE
2	No slot lands before eight or after six in that attendee’s own local time.	CODE
3	The explanation names at least one specific constraint it worked around.	JUDGE
4	If any calendar could not be read, the answer says so explicitly.	JUDGE
5	It proposes between two and four options, never one and never nine.	CODE

5 · YOUR FAILURES, NOT FREE SCORES — LESSON 7.5

The false green: library scores grade **text**, not your product. All four were right while it booked 7 am meetings.

HELPFUL	COHERENT	RELEVANT	FAITHFUL
0.92	0.95	0.91	0.89

- 12 Proposed a slot outside somebody’s working hours
- 9 Explanation named no actual constraint
- 6 Silently ignored a calendar it could not read
- 4 Offered more than four options

Good failure modes are **countable**, **traceable**, in **your words**. **Criteria drift** is learning: date every version; unchanged for three months means unused. Tripwires go on a **separate board**. **WHY YOU CARE:** a metric you did not choose is someone else’s opinion.